of an encrypted token, shown in figure 6.5. The five components of a JWE are:

1. The JWE header, which is very like the JWS header, but with two additional fields: `enc`, which specifies the encryption algorithm, and `zip`, which specifies an optional compression algorithm to be applied before encryption.
2. An optional encrypted key. This is used in some of the more complex encryption algorithms. It is empty for the direct symmetric encryption algorithm that is covered in this chapter.
3. The *initialization vector* or *nonce* used when encrypting the payload. Depending on the encryption method being used, this will be either a 12- or 16-byte random binary value that has been base64url-encoded.
4. The encrypted ciphertext.
5. The MAC authentication tag.

> **DEFINITION** An ***initialization vector*** (IV) or ***nonce*** (number-used-once) is a unique value that is provided to the cipher to ensure that ciphertext is always different even if the same message is encrypted more than once. The IV should be generated using a `java.security.SecureRandom` or other cryptographically-secure pseudorandom number generator (CSPRNG).[3] The IV doesn't need to be kept secret.